

Prescription for Sustainability – Environmental, Financial, and Performance Outcomes of Eco-Friendly Cold Chain Packaging

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Background

- Delivery of temperature-sensitive medications is a high-stakes, high-cost, and high-volume endeavor for:
 - patients
 - pharmacies
 - environment
- Conventional packaging relies heavily on:
 - expanded polystyrene (EPS)
 - plastic
- Purpose of packaging:
 - protect medications in range of climates and conditions
- Environmental footprint

Metric	Packaging	
	Conventional	Eco-friendly
CO ₂ (kg)	4.205	0.863
EPS waste (kg)	0.9	—
Cost	\$	

STUDY OBJECTIVE:

To characterize the potential advantages and disadvantages between conventional and eco-friendly packaging on common metrics of performance

Conventional Packaging

Cold-Chain Packaging



Methodology

- Design: prospective, multi-center, controlled trial; patient survey
- Setting: Ambulatory Care
- Population:
 - inclusion criteria:
 - medications delivered in cold chain packaging (January 2024 – present)
 - exclusion criteria:
 - medications delivered in ambient temperature packaging
- Primary outcome:
 - performance: temperature excursion (proportion of units, time to event)
- Secondary outcomes:
 - non-biodegradable waste volume
 - total consumer waste volume
 - recyclable waste volume
 - costs (e.g., acquisition, medication waste)
 - patient satisfaction, terminal destination for cold packaging (landfill), recycling (rate)
- Statistical Analysis Plan:
 - Fisher’s exact test (categorical data)
 - Student’s t-test (normally distributed data)
 - Mann-Whitney U (non-normally distributed data)
 - Multivariate linear and logistic regression
- Consider replacing headings with content, i.e., you may not need the outline headings

Discussion

- Value Proposition
 - sustainability initiatives align with operational and financial mission
 - perceived value with patients and stakeholders (e.g., trust and purchase intention)
 - requires transparency and accountability
- Environmental Impact:
 - *** # of cold-chain units and # of patients served → units per patient (for scale)
 - ***difference in CO2 emissions
- Future directions
 - improve delivery service logistics (e.g., lockers)
 - explore options to re-use packaging (e.g., exchange program)
 - evaluate multi-capacity coolers to optimize payload and reduce waste
 - utilize smaller coolers for small-volume shipments

Cold-chain medication packaging reimagined: greener, smarter, and more sustainable.

References

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1 = University of California – Davis Health, 2 = University of California San Francisco, 3 = University of California Los Angeles

Financial Disclosure(s): No one in a position to control the content of this educational activity has relevant financial relationships with ineligible companies